CHAPTER 9

Risk and Return: Portfolio Theory and Asset Pricing Models

Learning Objectives

After studying this chapter you should be able to:

- ✓ Calculate the portfolio risk for a 2-security case and an *n*-security case.
- ✓ Explain the concepts of efficient frontier and optimal portfolio.
- ✓ Discuss the capital market line and security market line.
- ✓ Explain the arbitrage pricing theory.
- ✓ Discuss multifactor models.

In our discussion of the basics of risk and return in the previous chapter, we learnt that:

- Much of the risk of a single security can be eliminated by diversification. So rational investors hold diversified portfolios.
- While diversification eliminates unsystematic risk (unique risk) it cannot eliminate systematic risk (market risk).
- The market risk of a stock is measured by its beta. According to the capital asset pricing model (CAPM), the expected return of a stock is linearly related to its beta.

Extending the previous discussion, this chapter provides an in-depth treatment of portfolio concepts and the CAPM. It also explores other asset pricing models.

9.1 PORTFOLIO RISK

Just as the risk of an individual security is measured by the variance (or standard deviation) of its return, the risk of a portfolio too is measured by the variance (or standard deviation) of its return.

Although the expected return on a portfolio is the weighted average of the expected returns on the individual securities in the portfolio, portfolio risk (measured by variance or standard deviation of returns) is not the weighted average of the risks of the individual securities in the portfolio (except when the returns from the securities are uncorrelated).

In symbols,

$$E(R_p) = \sum_{i=1}^{n} w_i E(R_i) \tag{9.1}$$

But

$$\sigma_p^2 \neq \sum w_i^2 \sigma_i^2 \tag{9.2}$$

Thanks to the inequality shown in Eq. (9.2), investors can achieve the benefit of risk reduction through diversification. Before we discuss how this can be accomplished let us first understand how comovements in security returns are measured.

Comovements in Security Returns

To develop the equation for calculating portfolio risk we need information on weighted individual security risks and weighted comovements between the returns on securities included in the portfolio.

Comovements between the returns on securities are measured by covariance (an absolute measure) and coefficient of correlation (a relative measure).

Covariance Covariance reflects the degree to which the returns on the two securities vary or change together. A positive covariance means that the returns on the two securities move in the same direction whereas a negative covariance implies that the returns on the two securities move in opposite direction. The covariance between the returns on any two securities *i* and *j* is calculated as follows:

$$\begin{aligned}\text{Cov}(R_i, R_j) = \; & p_1 [R_{i1} - E(R_i)] [R_{j1} - E(R_j)] \\ & + p_2 [R_{i2} - E(R_i)] [R_{j2} - E(R_j)] \\ & + \\ & . \\ & + p_n [R_{in} - E(R_i)] [R_{jn} - E(R_j)] \end{aligned} \tag{9.3}$$

where p_1, p_2 … p_n are the probabilities associated with states 1, … n, R_{i1}, … R_{in} are the returns on security i in states 1, … n, R_{j1}, … R_{jn} are the returns on security j in states 1, … n, and $E(R_i)$ and $E(R_j)$ are the expected returns on securities i and j.

Example The returns on securities 1 and 2 under five possible states of nature are given below:

State of nature	*Probability*	*Return on security 1*	*Return on security 2*
1	0.10	–10%	5%
2	0.30	15	12
3	0.30	18	19
4	0.20	22	15
5	0.10	27	12

The expected returns on securities 1 and 2 are as follows:

$$E(R_1) = 0.10\,(-10\%) + 0.30\,(15\%) + 0.30\,(18\%) + 0.20\,(22\%) + 0.10\,(27\%) = 16\%$$
$$E(R_2) = 0.10\,(5\%) + 0.30\,(12\%) + 0.30\,(19\%) + 0.20\,(15\%) + 0.10\,(12\%) = 14\%$$

The covariance between the returns on securities 1 and 2 is calculated below:

State of nature	Probability	Return on security 1	Deviation of the return on security 1 from its mean	Return on security 2	Deviation of the return on security 2 from its mean	Product of the deviations times probability
(1)	(2)	(3)	(4)	(5)	(6)	(2) × (4) × (6)
1	0.10	–10%	–26%	5%	–9%	23.4
2	0.30	15%	–1%	12%	–2%	0.6
3	0.30	18%	2%	19%	5%	3.0
4	0.20	22%	6%	15%	1%	1.2
5	0.10	27%	11%	12%	–2%	–2.2
						Sum = 26.0

Thus the covariance between the returns on the two securities is 26.0.

Coefficient of Correlation Covariance and correlation are conceptually analogous in the sense that both of them reflect the degree of comovement between two variables. Mathematically, they are related as follows:

$$\text{Cor}\,(R_i, R_j) \text{ or } \rho_{ij} = \frac{Cov\left(R_i, R_j\right)}{\sigma(R_i).\sigma\left(R_j\right)} \text{ or } \frac{\sigma_{ij}}{\sigma_i.\sigma_j} \tag{9.4}$$

$$\sigma_{ij} = \rho_{ij} \cdot \sigma_i \cdot \sigma_j \tag{9.5}$$

where Cor (R_i, R_j) = ρ_{ij} is the correlation coefficient between the returns on securities *i* and *j*, Cov (R_i, R_j) = σ_{ij} is the covariance between the returns on securities *i* and *j*, and $\sigma(R_i)$,$\sigma(R_j)$ = σ_i, σ_j are the standard deviations of the returns on securities *i* and *j*.

Thus the correlation coefficient is simply covariance divided by the product of standard deviations. The correlation coefficient can vary between −1.0 and +1.0. A value of -1.0 means perfect negative correlation or perfect comovement in the opposite direction; a value of 0 means no correlation or comovement whatsoever; a value of 1.0 means perfect correlation or perfect comovement in the same direction. Exhibit 9.1 portrays graphically various types of correlation relationships.

Calculation of Portfolio Risk

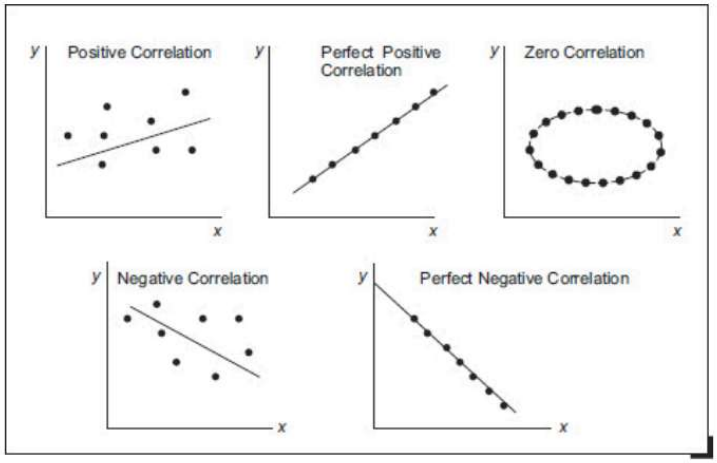
Now that we understand how to measure covariance and correlation, we will learn how to calculate portfolio risk. We will first look at the 2-security case and then generalise it to an *n*-security case.

Portfolio Risk: 2-Security Case The risk of a portfolio consisting of two securities is given by the following formula:

$$\sigma_p^2 = w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2\rho_{12}\sigma_1\sigma_2 \qquad (9.6)$$

where σ_p^2 is the variance of the portfolio return, w_1,w_2 are the weights of securities 1 and 2 in the portfolio, σ_1^2,σ_2^2 are the variances of the returns on securities 1 and 2, and $\rho_{12}\ \sigma_1\sigma_2$ is the covariance of the returns on securities 1 and 2.

Exhibit 9.1 **Graphical Portrayal of Various Types of Correlation Relationships**



In words, Eq. (9.6) says that the variance of the returns on a 2-security portfolio is the sum of the weighted variance of the returns on the two securities plus the weighted covariance between the returns on the two securities.

Example A portfolio consists of two securities, 1 and 2, in the proportions 0.6 and 0.4. The standard deviations of the returns on securities 1 and 2 are σ_1 = 10 percent and σ_2 = 16 percent. The coefficient of correlation between

the returns on securities 1 and 2 is 0.5. What is the standard deviation of the portfolio return?

$$\sigma_p = [0.6^2 \times 10^2 + 0.4^2 \times 16^2 + 2 \times 0.6 \times 0.4 \times 0.5 \times 10 \times 16]^{1/2}$$
$$= 10.7 \text{ percent}$$

The portfolio variance may be viewed as the sum of the four cells in a 2 × 2 matrix:

	Security 1	Security 2
Security 1	$w_1^2 \sigma_1^2$	$w_1 w_2 \rho_{12} \sigma_1 \sigma_2$
Security 2	$w_2 w_1 \rho_{21} \sigma_2 \sigma_1$	$w_2^2 \sigma_2^2$

The entry in the top left box, $w_1^2\ \sigma_1^2$, is the product of the square of the proportion invested in security 1 (w_1^2) and the variance of the returns on security 1 (σ_1^2); likewise, the entry in the bottom right box, $w_2^2\sigma_2^2$, is the product of the square of the proportion invested in security 2 (w_2^2) and the variance of the returns on security 2 (σ_2^2).

The entries in the other two boxes are the same. They are the product of the proportion invested in security 1 (w_1), the proportion invested in security 2 (w_2), and the covariance of the returns on the two securities (which may be expressed as $\rho_{12}\sigma_1\sigma_2$ or $\rho_{21}\sigma_2\sigma_1$).

Once you have filled in the four cells of the 2 × 2 matrix, you have to simply add the entries in these boxes to get the portfolio variance.

Portfolio Risk: *n*-Security Case The variance and standard deviation of the return of an *n*-security portfolio are:

$$\sigma_p^2 = \Sigma \Sigma\ w_i\ w_j\ \rho_{ij}\ \sigma_i\ \sigma_j \tag{9.7}$$

$$\sigma_p = [\Sigma \Sigma\ w_i\ w_j\ \rho_{ij}\ \sigma_i\ \sigma_j]^{1/2} \tag{9.8}$$

where σ_p^2 is the variance of portfolio return, σ_p is the standard deviation of portfolio return, w_i is the proportion of portfolio invested in security i, w_j is the proportion of portfolio invested in security j, ρ_{ij} is the coefficient of correlation between the returns on securities i and j, σ_i is the standard

deviation of return on security i, and σ_j is the standard deviation of return on security j.

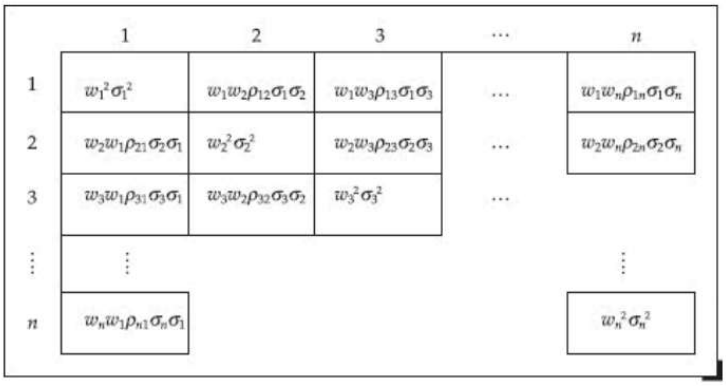
Example A portfolio consists of 3 securities, 1, 2, and 3. The proportions of these securities are: $w_1 = 0.5$, $w_2 = 0.3$, and $w_3 = 0.2$. The standard deviations of returns on these securities (in percentage terms) are: $\sigma_1 = 10$, $\sigma_2 = 15$, and $\sigma_3 = 20$. The correlation coefficients among security returns are: $\rho_{12} = 0.3$, $\rho_{13} = 0.5$, $\rho_{23} = 0.6$. What is the standard deviation of portfolio return?

$$\begin{aligned}\sigma_p &= [w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + w_3^2\sigma_3^2 + 2\,w_1w_2\rho_{12}\,\sigma_1\sigma_2 + 2\,w_1w_3\rho_{13}\sigma_1\sigma_3 + 2\,w_2w_3\rho_{23}\sigma_2\sigma_3]^{1/2} \\ &= [0.5^2\times 10^2 + 0.3^2\times 15^2 + 0.2^2\times 20^2 + 2\times 0.5\times 0.3\times 0.3\times 10\times 15 + 2\times 0.5\times 0.2 \\ &\quad \times 0.5\times 10\times 20 + 2\times 0.3\times 0.2\times 0.6\times 15\times 20]^{1/2} \\ &= 10.7 \text{ percent}\end{aligned}$$

The relationship embodied in Eq. (9.7) is not as complicated as it appears. It is the sum of n^2 terms found in the $n \times n$ matrix shown in Exhibit 9.2.

Notice that in Exhibit 9.2 there are n variance terms (the diagonal terms) and $n(n\text{-}1)$ covariance terms (the non-diagonal terms). If n is just two, there are two variance terms and two covariance terms. However, as n increases, the number of covariance terms is much larger than the number of variance terms. For example, when n is 10, there are 10 (that is n) variance terms and 90 (that is n (n-1)) covariance terms. Hence the variance of a well-diversified portfolio is largely determined by the covariance terms. If covariance terms are likely to be negative, it may be possible to get rid of risk almost wholly by resorting to diversification. Unfortunately, security prices tend to move together. This means that most covariance terms are positive. Hence, irrespective of how widely diversified a portfolio is, its risk does not fall below a certain level.

Exhibit 9.2 ***n* × *n* Matrix**



9.2 EFFICIENT FRONTIER AND OPTIMAL PORTFOLIO

Efficient Frontier

Now that we know how to calculate the risk and return of a portfolio, let us understand how the efficient frontier is delineated. We will first look at the 2-security case and then consider the *n*-security case.

Efficient Frontier for a Two Security Case Suppose an investor is evaluating two securities, A and B.

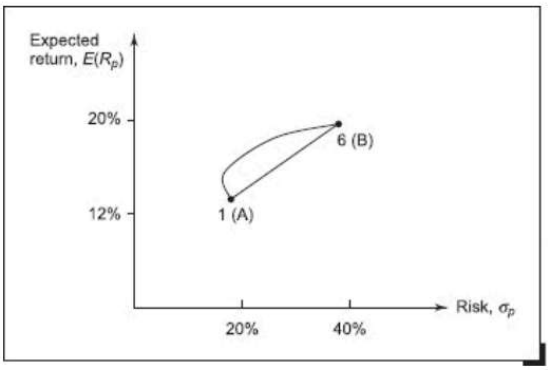
	Security A	*Security B*
Expected return	12%	20%
Standard deviation of return	20%	40%
Coefficient of correlation	−0.20	

The investor can combine securities A and B in a portfolio in a number of ways by simply changing the proportions of funds allocated to them. Some of the options available to him are shown below:

Portfolio	*Proportion of A* w_A	*Proportion of B* w_B	*Expected return* $E(R_p)$	*Standard deviation* σ_p
1 (A)	1.00	0.00	12.00%	20.00%
2	0.90	0.10	12.80%	17.64%
3	0.759	0.241	13.93%	16.27%
4	0.50	0.50	16.00%	20.49%
5	0.25	0.75	18.00%	29.41%
6 (B)	0.00	1.00	20.00%	40.00%

The six options described above are plotted graphically in Exhibit 9.3.

Exhibit 9.3 **Portfolio Options**



Efficient Frontier for the *n*-Security Case In a 2-security case, a curved line delineates all possible portfolios. In a multi-security case, the collection of all the possible portfolios is represented by the broken-egg shaped region, referred to as the feasible region, shown in Exhibit 9.4. Obviously, the number of possible portfolios in that region is virtually endless. However, the investor need not feel unduly overwhelmed by the bewildering range of possibilities shown in Exhibit 9.4 because what really matters to him is the northwest boundary of the feasible region which is defined as the thick dark line. Referred to as the efficient frontier, this boundary contains all the efficient portfolios. It may be useful to clarify here what exactly an efficient portfolio is. A portfolio is efficient if (and only if) there is no alternative with (i) the same $E(R_p)$ and a lower σ_p, (ii) the same σ_p and a higher $E(R_p)$, or (iii) a higher $E(R_p)$ and a lower σ_p. Thus, in Exhibit 9.4 while all the feasible portfolios are contained in the region *AFXMNO*, only the portfolios which lie along the boundary *AFX* are efficient. *AFX* represents the efficient frontier. All other portfolios are inefficient. A portfolio like *Z* is inefficient because portfolios like *B* and *D*, among others, dominate it. The efficient frontier is the same for all investors because portfolio theory is based on the assumption that investors have homogeneous expectations.

Optimal Portfolio

Once the efficient frontier is delineated, the next question is: What is the optimal portfolio for the investor? To determine the optimal portfolio on the efficient frontier, the investor's risk-return tradeoff must be known. Exhibit 9.5 presents two illustrative indifference curves which reflect risk-return tradeoff functions – note that all points lying on an indifference curve provide the same level of satisfaction. The indifference curves I_p and I_q represent the risk-return tradeoffs of two hypothetical investors, *P* and *Q*. Both *P* and *Q*, like most investors, are risk-averse. They want higher returns to bear more risk. *Q* is however more risk-averse than *P*. *Q* wants a higher expected return for bearing a given amount of risk as compared to *P*. In general, the steeper the slope of the indifference curve, the greater the degree of risk aversion.

Exhibit 9.4 **Feasible Region**

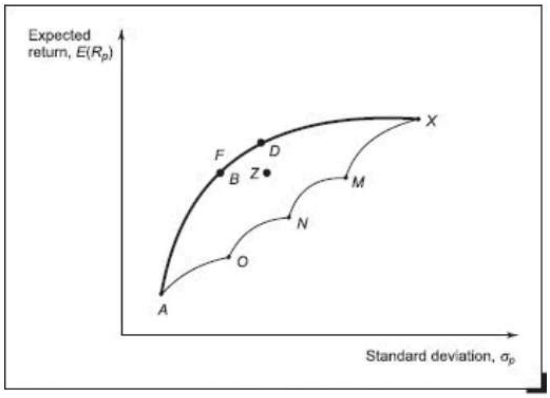
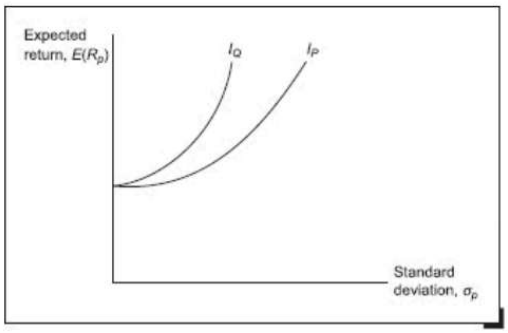
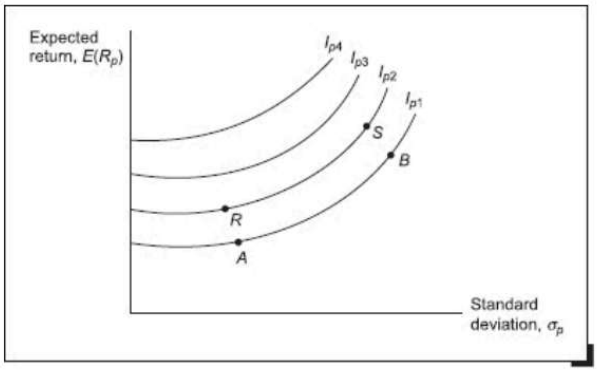


Exhibit 9.5 **Risk-Return Indifference Curves**



Each person has a map of indifference curves. Exhibit 9.6 shows the indifference map for *P*. In this figure, four risk-return indifference curves, I_{p1}, I_{p2}, I_{p3} and I_{p4} are shown. All the points lying on a given indifference curve offer the same level of satisfaction. For example, points *A* and *B*, which lie on the indifference curve I_{p1} offer the same level of satisfaction; likewise, points *R* and *S*, which lie on the indifference curve I_{p2}, offer the same level of satisfaction. The level of satisfaction increases as one moves leftward. The indifference curve I_{p2} represents a higher level of satisfaction as compared to the indifference curve I_{p1}; the indifference curve I_{p4} represents a higher level of satisfaction when compared to the indifference curve I_{p3}; and so on.

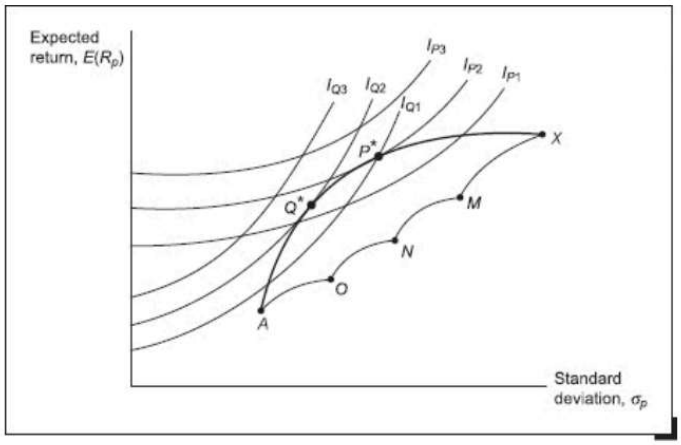
Exhibit 9.6 **Utility Indifference Curves**



Given the efficient frontier and the risk-return indifference curves, the optimal portfolio is found at the point of tangency between the efficient

frontier and a utility indifference curve. This point represents the highest level of utility the investor can reach. In Exhibit 9.7 two investors *P* and *Q*, confronted with the same efficient frontier, but having different utility indifferences curves (I_{p1}, I_{p2} and I_{p3} for *P* and I_{q1}, I_{q2} and I_{q3} for *Q*) are shown to achieve their highest utilities at points *P** and *Q** respectively.

Exhibit 9.7 **Optimal Portfolio**



Optimal Portfolio with Lending and Borrowing Let us introduce yet another opportunity. Suppose that the investor can also lend and borrow money at a risk-free rate R_f percent which is shown in Exhibit 9.8. If he lends a portion of his funds at R_f and invests the balance in *S (S* is the point on the efficient frontier of risky portfolios where the straight line emanating from R_f is tangential to the efficient frontier or risky portfolios), he can obtain any combination of risk and return along the line that connects R_f and *S*. Further, if he borrows some money and invests it along with his own funds, he can reach a point like *G*, beyond *S*, as shown in Exhibit 9.8. Thus, with the opportunity of lending and borrowing, the efficient frontier changes. It is no longer *AFX*. Rather, it becomes R_f *SG* because R_f *SG*, as is clear from Exhibit 9.8, dominates *AFX*. For every point on *AFX*, there is at least one point on R_f *SG* which is superior to the point on *AFX*. For example, compared to *C* on *AFX*, *D* on R_f *SG* offers a higher expected return for the same standard deviation; likewise, compared to *B* on *AFX*, *E* on R_f *SG* offers the same expected return with a lower standard deviation; and so on.

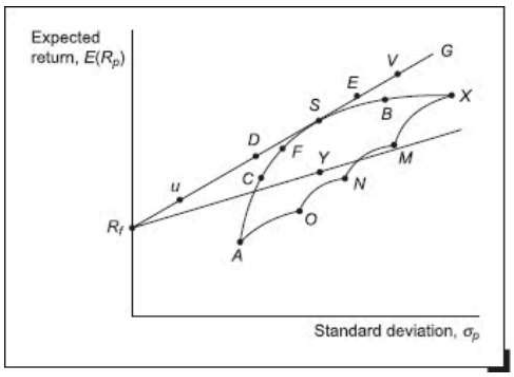
Since R_f SG dominates AFX, every investor would do well to choose some combination of R_f and S. A conservative investor may choose a point like U, whereas an aggressive investor may choose a point like V. However, note that both of them choose some combination of R_f and S. While the conservative investor weights R_f more in his portfolio, the aggressive investor weights S more in his portfolio (in fact, in his portfolio, the weight assigned to R_f is negative and the weight assigned to S is more than 1).

Thus, the task of portfolio selection can be separated into two steps.

- Identification of S, the optimal portfolio of risky securities.
- Choice of a combination of R_f and S depending on one's risk attitude.

This is the import of the celebrated **separation theorem**, first proposed by James Tobin, a Nobel laureate in Economics.

Exhibit 9.8 **Lending and Borrowing Opportunity**



9.3 CAPITAL ASSET PRICING MODEL

Harry Markowitz, a Nobel laureate in Economics, developed an approach that helps an investor to achieve his optimal portfolio position. Hence, portfolio theory, in essence, has a normative character as it prescribes what a rational investor should do.

William Sharpe and others asked the follow-up question: If rational investors follow the Markowitzian prescription, what kind of relationship exists between risk and return? Essentially, the capital asset pricing model (CAPM) developed by them is an exercise in positive economics. It is concerned with two key questions:

- What is the relationship between risk and return for an efficient portfolio?
- What is the relationship between risk and return for an individual security?

The CAPM, in essence, predicts the relationship between the risk of an asset and its expected return. This relationship is very useful in two important ways. First, it produces a benchmark for evaluating various investments. For example, when we are analysing a security we are interested in knowing whether the expected return from it is in line with its fair return as per the CAPM. Second, it helps us to make an informed guess about the return that can be expected from an asset that has not yet been traded in the market. For example, how should a firm price its initial public offering of stock?

Although the empirical evidence on the CAPM is mixed, it is widely used because of the valuable insight it offers and its accuracy is deemed satisfactory for most practical applications. No wonder, the CAPM is a centrepiece of modern financial economics and William Sharpe, its principal originator, was awarded the Nobel Prize in Economics.

The CAPM is based on the following assumptions:

- Investors are risk averse.
- Security returns are normally distributed.
- The utility function of investors is quadratic.
- Investors have homogeneous expectations – they have identical subjective estimates of the means, variances, and covariances among returns.
- Investors can borrow and lend freely at a riskless rate of interest.

- The market is perfect: there are no taxes; there are no transactions costs; securities are completely divisible; the market is competitive.
- The quantity of risky securities in the market is given.

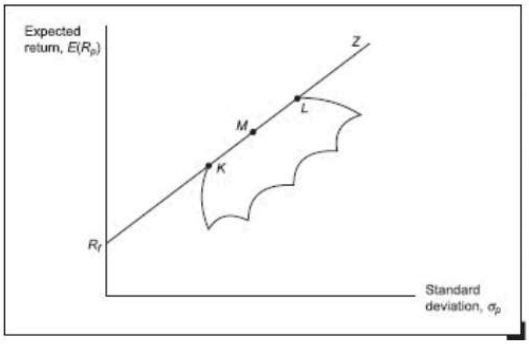
Looking at these assumptions, one may feel that the CAPM is unrealistic. However, the value of a model depends not on the realism of its assumptions, but on the validity of its conclusions. Extensive empirical analysis suggests that there is merit in the CAPM.

Capital Market Line

In our discussion of portfolio theory, we learnt that rational investors would choose a combination of R_f and S (S represents the point on the efficient frontier of risky portfolios where the straight line emanating from R_f is tangential to the efficient frontier). If all investors attempt to purchase the securities in S and ignore securities not included in S, prices of securities would be revised. On the one hand, prices of securities included in S would rise and hence their expected returns will fall. This would shift S, along with other points which share securities with S, downward. On the other hand, prices of securities not included in S will fall, leading to an increase in their expected return. Consequently, points representing portfolios in which these securities are included will shift upward. As this process continues, the efficient frontier of risky securities will flatten as shown in Exhibit 9.9. Finally, the set of prices reached would be such that every security will enter at least one portfolio on the linear segment KML. Of course, the market portfolio would itself be a point on that linear segment.

Portfolios which have returns that are perfectly positively correlated with the market portfolio are referred to as efficient portfolios. Obviously, these are portfolios that lie on the linear segment.

Exhibit 9.9 **Adjustment of the Efficient Frontier**



For efficient portfolios (which includes the market portfolio) the relationship between risk and return is depicted by the straight line $R_f MZ$. The equation for this line, called the capital market line (CML), is:

$$E(R_j) = R_f + \lambda\sigma_j \qquad (9.9)$$

where $E(R_j)$ is the expected return on portfolio *j*, R_f is the risk-free rate, λ is the slope of the capital market line, and σ_j is the standard deviation of portfolio *j*.

Given that the market portfolio has an expected return of $E(R_M)$ and standard deviation of σ_M, the slope of the CML can be obtained as follows:

$$\lambda = \frac{E(R_M) - R_f}{\sigma_M} \tag{9.10}$$

where λ, the slope of the CML, may be regarded as the "price of risk" in the market.

Security Market Line

As discussed above, as far as efficient portfolios are concerned, there is a simple linear relationship between expected return and standard deviation. What about individual securities and inefficient portfolios? Typically, the expected return and standard deviation for individual securities will be below the CML, reflecting the inefficiency of undiversified holdings. Further, such points would be found throughout the feasible region with no well-defined relationship between their expected return and standard deviation. However, there is a linear relationship between their expected return and their covariance with the market portfolio. This relationship, called the security market line (SML), is as follows:

$$E(R_i) = R_f + \left[\frac{E(R_M) - R_f}{\sigma^2_M}\right]\sigma_{iM} \tag{9.11}$$

where $E(R_i)$ is the expected return on security *i,* R_f is the risk-free return, $E(R_M)$ is the expected return on market portfolio, σ^2_M is the variance of return on market portfolio, and σ_{iM} is the covariance of return between security *i* and market portfolio.

In words, the SML relationship says:

Expected return on security *i* = Risk-free return + (Price per unit of risk) Risk

The price per unit of risk is: $\frac{E(R_M) - R_f}{\sigma^2_M}$

The measure of risk is: σ_{iM}

In Eq. (9.11), the risk of a security is expressed in terms of its covariance with the market portfolio, $\sigma_{iM.}$

Can we find a standardised measure of risk? Fortunately we can find a standardised measure of systematic risk, popularly called beta (β), by taking advantage of the relationship:

$$\beta_i = \frac{\sigma_{iM}}{\sigma^2_M} \tag{9.12}$$

which reflects the slope of a linear regression relationship in which the return on security *i* is regressed on the return on the market portfolio.

Thus, the SML is popularly expressed as

$$E(R_i) = R_f + [E(R_M) - R_f]\,\beta_i \tag{9.13}$$

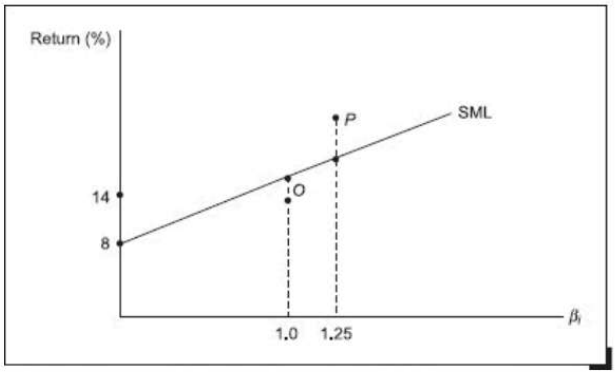
In words, the SML relationship says:

Expected return on security *i* = Risk-free return + Market risk premium × Beta of security *i*

The SML which reflects the expected return-beta relationship is shown in Exhibit 9.10. Note that the slope of the SML is the market risk premium.

Assets which are fairly priced plot exactly on the SML. Under-priced securities plot above the SML, whereas over-priced securities plot below the SML. The difference between the actually expected return on a security and its fair return as per the SML is called the security's **alpha**, denoted by α.

Exhibit 9.10 **The Security Market Line**



Empirical Evidence on CAPM

Beta, a product of academic research, was initially viewed with disdain and suspicion by the investment community. However, it was gradually accepted, as the initial empirical evidence supported it. The earlier resistance turned into enthusiasm. Beta indeed became very fashionable in the 1970s and the investment industry in the U.S. began manufacturing and supplying beta on a large scale. Along with the spread of the beta cult, the capital asset pricing and its various extensions have been subjected to rigorous scrutiny and testing.

According to the capital asset pricing model, the expected return on a security is:

$$E(R_i) = R_f + \beta_i [E(R_M) - R_f] \tag{9.14}$$

The ideal way to test the CAPM would be to observe investors' expectations of betas and expected returns on individual securities and the market portfolio and then compare the expected return on each security with its return predicted by the CAPM. Unfortunately, this procedure is not practical since information on investor expectations is very sketchy.

In practice, researchers have tested the CAPM using *ex post data*, rather than *ex ante* data. They have examined the relationship between the security beta and realised return.

$$\bar{R}_i = v_o + v_1 \beta_i + e_i \tag{9.15}$$

where $\bar{R}_i$ is the realised return on security i, v_o is the intercept, β_i is the estimated beta of security i, and v_1 is the slope coefficient.

If the CAPM holds:

- The relationship should be linear. This means that terms like β_i^2, if substituted for β_i, should not yield better explanatory power.
- v_o, the intercept, should not be significantly different from the risk-free rate, R_f
- v_1, the slope coefficient, should not be significantly different from $\overline{R_M - R_f}$
- No other factors, such as company size or total variance, should affect R_i
- The model should explain a significant portion of variation in returns among securities.

Numerous empirical studies have been conducted to test the CAPM. Without going into the details of the individual studies, let us note the following general conclusions that emerge from these studies.

- The relation appears to be linear.
- In general v_o is greater than the risk-free rate and v_1 is less than $\overline{R_M - R_f}$. This means that the actual relationship between risk (as measured by beta) and return is flatter than what the CAPM says.
- In addition to beta, some other factors, such as standard deviation of returns, price – earnings multiple and company size, too have a bearing on return.
- Beta does not explain a very high percentage of the variance in return among securities.

While reviewing the empirical evidence, bear in mind two important problems. First, the studies use historical returns as proxies for expectations. This assumes that the expected returns will be the same as the realised returns. Second, the studies use a market index as a proxy for the market portfolio. Richard Roll has argued persuasively that since the 'true' market portfolio (which in principle must include all assets – financial, real, as well as human – and not just equity stocks), cannot be measured, the CAPM cannot be tested.

Notwithstanding the problems mentioned above, the CAPM is the most widely used risk -return model. Its popularity may be attributed to the following factors:

- Beta can be estimated with short data; risk-free rate can be easily established; the market risk premium can be computed, as information is available for a long period.
- Some objective estimate of risk premium is better than a completely subjective estimate or no estimate.
- CAPM is a simple and intuitively appealing risk-return model. Its basic message that diversifiable risk does not matter is accepted by nearly every one.
- While there are plausible alternative risk measures, no consensus has emerged on what course to plot if beta is abandoned. As Brealey and Myers say: "So the capital asset pricing model survives not from a lack of competition but from a surfeit."

The situation perhaps may change as additional evidence is gathered in favour of arbitrage pricing model and operational guidelines for applying

that model are developed further. As of now, however, the CAPM appears to be the model of choice in practice.

9.4 ARBITRAGE PRICING THEORY AND MULTIFACTOR MODELS

Arbitrage Pricing Theory

While the CAPM represents a seminal contribution to the field of finance, many empirical studies have pointed towards its deficiencies in explaining the relationship between risk and return.

A key challenge to the CAPM came from a set of studies that have suggested that it is possible to rely on certain firm or security characteristics and earn superior returns even after adjustment for risk as measured by beta. Examples: Banz found that small cap stocks outperformed large cap stocks on a risk-adjusted basis; Basu found that low *P/E* stocks outperformed high *P/E* stocks, after adjustment for risk; more recently, Fama and French documented that "value stocks" (stocks with high book-to-market price ratios) generated larger returns than "growth stocks" (stocks with low book-to-market ratios), on a risk-adjusted basis.

In an efficient market such return differentials should not exist. Does it mean that the markets are not particularly efficient for long periods of time? Or, does it mean that the markets are efficient but a single-factor model such as the CAPM does not capture risk adequately?

Since it is unlikely that markets are inefficient for extended periods of time, financial economists began looking for alternative risk-return models, beyond the CAPM. In particular, they asked: What happens if many factors are required to specify the equilibrium relationship between risk and return? Stephen Ross developed the Arbitrage Pricing Theory (APT) that can include any number of risk factors. The APT is based on complex mathematical and statistical analysis that is beyond the scope of this text. Although the APT is theoretically superior to the CAPM, its practical applications have been rather limited. Perhaps its usage may increase in future, so you should at least have an intuitive idea about it.

According to the SML, a stock's expected rate of return is equal to the risk-free rate plus a risk premium which is the market risk premium times the stock's beta coefficient. Thus,

$$E(R_i) = R_f + (E(R_M) - R_f)\beta_i \tag{9.16}$$

The historical realised return, R_i, however, will generally be different from the expected return. It may be expressed as follows:

$$R_i = E(R_i) + (R_M - E(R_M))\beta_i + e_i \tag{9.17}$$

Thus, the realised return, R_i, is equal to the expected return, $E(R_i)$, plus a positive or negative element, $R_M - E(R_M)\beta_i$ which depends jointly on whether the market performed better or worse than was expected and the stock's beta, plus a random error term, e_i.

Note that Eq. (9.17) represents the **return generating process** whereas Eq. (9.16) represents the **equilibrium risk-return relationship** as per the CAPM.

Now, the market's realised return, R_M, is, in turn, influenced by a number of factors such as GDP, inflation level, corporate profitability, investor sentiment, and so on. Further, these factors affect different groups of stocks in different ways. So, one could specify the expected and realised returns on individual stocks as a function of different fundamental economic factors.

The expected rate of return on a stock, according to the APT, is:

$$E(R_i) = R_f + (E(R_1) - R_f)\beta_{i1} + \cdots + (E(R_j) - R_f)\beta_{ij} \tag{9.18}$$

In this equation $E(R_j)$ is the expected rate of return on a portfolio that is sensitive only to the jth economic factor ($\beta_{Pj} = 1.00$), but has zero sensitivity to every other factor. For example, $E(R_2) - R_f$ is the risk premium on a portfolio with ($\beta_{P2} = 1.0$) and all other $\beta_{Pj} = 0$.

Note that Eq. (9.18) is the **equilibrium risk-return relationship** in APT. It is equivalent of the CAPM's Security Market Line. The APT, unlike the CAPM, permits a stock's return to be a function of multiple factors.

The realised rate of return equation in APT, which represents the **return generating function** is as follows:

$$R_i = E(R_i) + (F_1 - E(F_1))\beta_{i1} + \cdots + (F_j - E(F_j))\beta_{ij} + e_i \tag{9.19}$$

where R_i is the realised rate of return on stock i, $E(R_i)$ is the expected rate of return on stock i, F_j is the realised value of economic factor j, $E(F_j)$ is the expected value of economic factor j, β_{ij} is the sensitivity of stock i to economic factor j, and e_i is a random term that reflects the effect of unique events on stock i.

Thus, the realised return on any stock is equal to the expected return on the stock *plus* increments or decrements that depend on unexpected

changes in fundamental economic factors times the sensitivity of the stock to these changes *plus* a random term.

To illustrate the APT, assume that there are three risk factors that determine the returns on all stocks: inflation, gross domestic production, and the aggregate degree of risk aversion (measured as the spread between the yields of low-grade bonds and treasury bonds). Further, assume the following:

- R_f, the risk-free rate, is 8%.
- $E(R_1)$, the expected rate of return on a portfolio with unit sensitivity to inflation (β_{P1} = 1.0), and zero sensitivities to GDP and degree of risk aversion, is 14%.
- $E(R_2)$, the expected rate of return on a portfolio with unit sensitivity to GDP (β_{P2} = 1.0), and zero sensitivities to inflation and degree of risk aversion, is 12%.
- $E(R_3)$, the expected rate of return on a portfolio with unit sensitivity to degree of risk aversion (β_{P3} = 1.0), and zero sensitivities to inflation and GDP, is 7%.

Now consider two stocks, A and B, with the following factor sensitivities.

	Sensitivity to Factor		
	Inflation	GDP	Degree of Risk Aversion
Stock A	0.8	1.3	–0.6
Stock B	0.6	0.7	–0.8

The expected rate of return on stocks A and B would be as follows:

Stock A: $E(R_A) = 8\% + (14\% - 8\%)\ 0.8 + (12\% - 8\%)\ 1.3 + (7\% - 8\%)\ (-0.6) = 18.6\%$

Stock B: $E(R_B) = 8\% + (14\% - 8\%)\ 0.6 + (12\% - 8\%)\ 0.7 + (7\% - 8\%)\ (-0.8) = 15.2\%$

APT is theoretically superior to CAPM in several ways:

(i) APT allows for several economic factors to determine individual stock returns, whereas the CAPM assumes that the effect of all factors, barring those unique to the firm, is captured in a single measure, viz., the volatility of the stock in relation to the market portfolio.

(ii) APT makes fewer assumptions unlike the CAPM; it does not require the following assumptions:

- the utility functions of investors are quadratic;
- security returns are normally distributed;
- all investors hold the market portfolio;
- the market portfolio comprising of all risky assets is mean- variance efficient.

The APT only assumes that the capital market is perfectly competitive and that investors always prefer more wealth to less wealth with certainty.

Comparison of CAPM and APT

	CAPM	*APT*
Nature of relation	Linear	Linear
Number of risk factors	1	j
Factor risk premium	$[E(R_M) - R_f]$	$[E(R_i) - R_f]$
Factor risk sensitivity	β_i	β_{ij}
Zero-beta return	R_f	R_f

Empirical Evidence Unlike the CAPM, the APT does not specify *a priori* what the underlying risk factors are. So, a test of APT calls for first discovering the basic risk factors by employing multivariate techniques like factor analysis and then examining whether these basic risk factors correspond to some economic or behavioural variables.

Empirical studies of this kind so far suggest that there is hardly any consistency in terms of (i) the number of basic factors, (ii) the interpretation that may be put on these factors (typically, the factors identified are artificial constructs representing several economic variables), and (iii) the stability of these factors from test to test.

Multifactor Models

Given the practical difficulties in using the above approach, researchers have followed a different approach that captures the essence of the APT. In this approach, the researcher chooses *a priori* the exact number and identity of risk factors and specifies the **multifactor model** of the following kind.

$$R_{it} = a_i + [\, b_{i1} F_{1t} + b_{i2} F_{2t} + \ldots + b_{ik} F_{kt} \,] + e_{it} \tag{9.20}$$

where R_{it} is the return on security i in period t, F_{jt} is the return associated with the jth risk factor in period t, b_{it} is the sensitivity of return on security i to factor j.

The advantage of a factor model like this is that the researcher can specify the risk factors; the disadvantage of such a model is that there is very little theory to guide it. Hence, developing a useful factor model is as much an art as science.

The variety of multifactor models employed in practice may be divided into two broad categories: macro-economic based risk factor models and micro-economic based risk factor models.

Macroeconomic Based Risk Factor Models These models consider risk factors that are macroeconomic in nature. Typical of such approach is the following model proposed by Chen, Roll, and Ross in their classic paper, "Economic Forces and the Stock Market," published in the April 1986 issue of *Journal of Business.*

$$R_{it} = a_i + b_{i1} R_{mt} + b_{i2} MP_t + b_{i3} DEI_t + b_{i4} UI_t + b_5 UPR_t + b_{i6} UTS_t + e_{it} \tag{9.21}$$

where R_m is the return on a value weighted index of NYSE – listed stocks, *MP* is the monthly growth rate in the US industrial production, *DEI* is the change in inflation, measured by the US consumer price index, *UI* is the difference between actual and expected levels of inflation, *UPR* is the unanticipated change in the bond credit spread (Baa yield – RFR), and *UTS* is the unanticipated term structure shift (long term RFR – short term RFR).

Microeconomic Based Risk Factor Models Instead of specifying risk in macroeconomic terms, you can delineate risk in microeconomic terms. Typical of this approach is the following model proposed by Fama and French in their celebrated paper "Common Risk Factors in the Returns

on Stocks and Bonds," published in the January 1993 issue of the *Journal of Financial Economics:*

$$(R_{it} - RFR_t) = \alpha_i + b_{i1}(R_{mt} - RFR_t) + b_{i2}SMB_t + b_{i3}HML_t + e_{it} \quad (9.22)$$

In this model, in addition to $(R_{mt} - RFR_t)$, the excess return on a stock market portfolio, there are two other microeconomic risk factors: SMB_t and HML_t. SMB_t (i.e., small minus big) is the return on a portfolio of small capitalisation stocks less the return on a portfolio of large capitalisation stocks and HML_t (i.e., high minus low) is the return on a portfolio of stocks with high ratios of book-to-market values less the return on a portfolio of low book-to-market value stocks.

In this model, *SMB* is intended to capture the risk associated with firm size while *HML* is meant to reflect risk differentials associated with "value" (i.e., high book-to-market ratio) and "growth" (i.e., low book-to-market ratio) stocks.

Choice of Model Compared to the CAPM, multifactor models explain past returns better. However, when it comes to predicting expected returns, the evidence is ambiguous. It appears that the gains from having multiple factors are offset by the errors committed in estimating the factor loadings and the factor betas. The widespread use of the CAPM stems from its intuitive appeal and its simplicity.

SUMMARY

- **Covariance** and **correlation** are conceptually analogous in the sense that both of them reflect the degree of **comovement** between the variables.
- The **standard deviation** of a two-security portfolio is:

$$\sigma_p = [w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2\rho_{12}\sigma_1\sigma_2]^{1/2}$$

- The **standard deviation** of an *n*-security portfolio is:

$$\sigma_p = [\Sigma\Sigma w_i w_j \rho_{ij} \sigma_i \sigma_j]^{1/2}$$

- A portfolio is **efficient** if (and only if) there is no alternative with (i) the same $E(R_p)$ and a lower σ_p or (ii) the same σ_p and a higher $E(R_p)$ or (iii) a higher $E(R_p)$ and a lower σ_p.
- Each person has a map of **indifference curves**. All the points lying on a given indifference curve offer the same level of satisfaction.
- Given the efficient frontier and the risk-return indifference curves, the **optimal portfolio** is found at the tangency between the efficient frontier and a utility

indifference curve.

- If lending and borrowing can be done at the risk-free rate, the efficient frontier is simply the straight line from the risk-free rate which is tangential to the broken-egg shaped feasible region.
- The **capital market line** equation is: $E(R_i) = R_f + \lambda\sigma_j$
- The **security market line** equation is: $E(R_i) = R_f + [E(R_M) - R_f]\,\beta_i$
- The empirical evidence in favour of the **Capital Asset Pricing Model (CAPM)** is mixed. Yet, the CAPM is the most widely used risk return model because it is simple and intuitively appealing and its basic message that diversifiable risk does not matter is accepted by nearly everyone.
- The **Arbitrage Pricing Theory (APT)** assumes that the return on any stock is linearly related to a set of systematic factors:

$$R_i = E(R_i) + (F_1 - E(F_1))\beta_{i1} + \cdots + (F_j - E(F_j))\beta_{ij} + e_i$$

- The equilibrium relationship according to APT is

$$E(R_i) = R_f + (E(R_1) - R_f)\beta_{i1} + \cdots + (E(R_j) - R_f)\beta_{ij}$$

- The APT does not **a priori** specify what the underlying risk factors are. A **multifactor model**, on the other hand, chooses **a priori** the exact number and identity of risk factors.
- The variety of multifactor models employed in practice fall into two categories: **macro-economic based risk factor models** and **micro-economic based risk factor models**.

QUESTIONS

1. What is the risk of a 2-security portfolio? *n*-security portfolio?
2. What is covariance? coefficient of correlation?
3. What is an efficient portfolio?
4. Explain the nature of a risk-return indifference curve.
5. What is the efficient frontier, when investors can lend or borrow at the risk-free rate?
6. State the assumptions underlying the CAPM.
7. What is the capital market line? Security market Line?
8. What is the empirical evidence on the CAPM?
9. Define the return-generating process and the equilibrium risk-return relationship as per the APT.
10. What is a multifactor model? Describe the types of multifactor models used in practice?

SOLVED PROBLEMS

9.1 The returns on two assets under four possible states of nature are given below:

State of nature	*Probability*	*Return on asset 1*	*Return on asset 2*
1	0.10	5%	0%
2	0.30	10%	8%
3	0.50	15%	18%
4	0.10	20%	26%

(a) What is the standard deviation of the return on asset 1? asset 2?
(b) What is the covariance between the returns on assets 1 and 2?
(c) What is the coefficient of correlation between the returns on assets 1 and 2?

Solution:

(a) The expected return on assets 1 and 2 is:

$$E(R_1) = 0.1\,(5\%) + 0.3\,(10\%) + 0.5\,(15\%) + 0.1\,(20\%)$$
$$= 13\%$$
$$E(R_2) = 0.1\,(0\%) + 0.3\,(8\%) + 0.5\,(18\%) + 0.1\,(26\%)$$
$$= 14\%$$

The standard deviation of the returns on assets 1 and 2 are

$$\sigma_1 = [0.1\,(5-13)^2 + 0.3\,(10-13)^2 + 0.5\,(15-13)^2 + 0.1\,(20-13)^2]^{1/2} = 4\%$$
$$\sigma_2 = [0.1\,(0-14)^2 + 0.3\,(8-14)^2 + 0.5\,(18-14)^2 + 0.1\,(26-14)^2]^{1/2}$$
$$= [19.6 + 10.8 + 8 + 14.4]^{1/2} = 7.27\%$$

(b) The covariance between the returns on assets 1 and 2 is calculated below

State of nature	*Probability*	*Return of asset 1*	*Deviation of the return on asset 1 from its mean*	*Return on asset 2*	*Deviation of the return on asset 2 from its mean*	*Product of deviations times Probability*
(1)	*(2)*	*(3)*	*(4)*	*(5)*	*(6)*	*(7)*
1	0.10	5%	−8%	0%	−14%	11.2
2	0.30	10%	−3%	8%	−6%	5.4
3	0.50	15%	2%	18%	4%	4
4	0.10	20%	7%	26%	12%	8.4
						Sum = 29.0

Thus the covariance between the returns of the two assets is 29.0

(c) The coefficient of correlation between the returns on assets 1 and 2 is:

$$\frac{\text{Covariance}_{12}}{\sigma_1 \times \sigma_2} = \frac{29}{4 \times 7.27} = 0.997$$

9.2 A portfolio consists of 3 securities, 1, 2, and 3. The proportions of these securities are: $w_1 = 0.3$, $w_2 = 0.5$, and $w_3 = 0.2$. The standard deviations of returns on these securities (in percentage terms) are: $\sigma_1 = 6$, $\sigma_2 = 9$, and $\sigma_3 = 10$. The correlation coefficients among security returns are $\rho_{12} = 0.4$, $\rho_{13} = 0.6$, $\rho_{23} = 0.7$. What is the standard deviation of portfolio return?

Solution:

$$\sigma_p = [w^2_1\sigma^2_1 + w^2_2\sigma^2_2 + w^2_3\sigma^2_3 + 2w_1w_2\rho_{12}\sigma_1\sigma_2 + 2w_1w_3\rho_{13}\sigma_1\sigma_3 + 2w_2w_3\rho_{23}\sigma_2\sigma_3]^{1/2}$$
$$= [0.3^2\times6^2 + 0.5^2\times9^2 + 0.2^2\times10^2 + 2\times0.3\times0.5\times0.4\times6\times9 + 2\times0.3\times0.2\times0.6\times 6\times10 + 2\times0.5\times0.2\times0.7\times9\times10]^{1/2}$$
$$= [3.24 + 20.25 + 4 + 6.48 + 4.32 + 12.6]^{1/2} = 7.13\%$$

9.3 The following information is available.

	Stock A	*Stock B*
Expected return	16%	12%
Standard deviation	15%	8%
Coefficient of correlation		0.60

(a) What is the covariance between stocks *A* and *B*?
(b) What is the expected return and risk of a portfolio in which *A* and *B* have weights of 0.6 and 0.4?

Solution:

(a) $\text{Covariance}(A,B) = \rho_{AB}\times\sigma_A\times\sigma_B$
$= 0.60\times15\times8 = 72$

(b) Expected return $= 0.6\times16 + 0.4\times12 = 14.4\%$
Risk (standard deviation) $= [w_A{}^2\sigma_A{}^2 + w_B{}^2\sigma_B{}^2 + 2\,\text{Cov}(A,B)]^{1/2}$
$= [0.6^2\times225 + 0.4^2\times64 + 272]^{1/2}$
$= 15.34\%$

9.4 The following table gives an analyst's expected return on two stocks for particular market returns:

Market Return	*Aggressive Stock*	Defensive Stock
6%	2%	8%
20	30	16

(a) What are the betas of the two stocks?
(b) What is the expected return on each stock if the market return is equally likely to be 6% or 20%?
(c) If the risk-free rate is 7% and the market return is equally likely to be 6% or 20% what is the SML?
(d) What are the alphas of the two stocks?

Solution:

(a) The betas of the two stocks are:

$$\text{Aggressive stock} = \frac{30\% - 2\%}{20\% - 6\%} = 2$$

$$\text{Defensive stock} = \frac{16\% - 8\%}{20\% - 6\%} = 0.571$$

(b) The expected return on the two stocks are:

$$\text{Aggressive stock} : 0.5 \times 2\% + 0.5 \times 30\% = 16\%$$
$$\text{Defensive stock} : 0.5 \times 8\% + 0.5 \times 16\% = 12\%$$

(c) The expected return on the market portfolio is:

$$0.5 \times 6\% + 0.5 \times 20\% = 13\%$$

Since the risk-free rate is 7%, the market risk premium is: 13% − 7% = 6%

So, the SML is:

Required return$_i$ = 7% + β_i × 6%

(d) The alphas of the two stocks are calculated below

Stock A

Expected return = 16%

Beta = 2

Required return = 7% + 2 × 6% = 19%

Alpha = 16% − 19% = − 3%

Stock B

Expected return = 12%

Beta = 0.571

Required return = 7% + 0.571 × 6% = 10.426%

Alpha = 12% − 10.426% = 1.574%

PROBLEMS

9.1 Correlation The returns on two assets under four possible states of nature are given below:

State of nature	*Probability*	*Return on asset* 1	*Return on asset* 2
1	0.20	–5%	10%
2	0.30	15%	12%
3	0.40	18%	14%
4	0.10	22%	18%

(a) What is the standard deviation of the return on asset 1 and on asset 2?

(b) What is the covariance between the returns on assets 1 and 2?

(c) What is the coefficient of correlation between the returns on assets 1 and 2?

9.2 Portfolio Return The returns on 4 stocks, *A, B, C*, and *D* over a period of 6 years have been as follows:

	1	2	3	4	5	6
A	10%	12%	−8%	15%	−2%	20%
B	8%	4%	15%	12%	10%	6%
C	7%	8%	12%	9%	6%	12%
D	9%	9%	11%	4%	8%	16%

Calculate the return on:

(a) portfolio of one stock at a time
(b) portfolios of two stocks at a time
(c) portfolios of three stocks at a time
(d) a portfolio of all the four stocks

Assume equiproportional investment.

9.3 Portfolio Standard Deviation A portfolio consists of 4 securities, 1, 2, 3, and 4. The proportions of these securities are: $w_1 = 0.2$, $w_2 = 0.3$, $w_3 = 0.4$, and $w_4 = 0.1$. The standard deviations of returns on these securities (in percentage terms) are : $\sigma_1 = 4$, $\sigma_2 = 8$, $\sigma_3 = 20$, and $\sigma_4 = 10$. The correlation coefficients among security returns are: $\rho_{12} = 0.3$, $\rho_{13} = 0.5$, $\rho_{14} = 0.2$, $\rho_{23} = 0.6$, $\rho_{24} = 0.8$, and $\rho_{34} = 0.4$. What is the standard deviation of portfolio return?

9.4 Correlation Stock P and Q will provide the following returns under different states of the economy.

State of Economy	*Probability*	*Return on Stock P*	*Return on Stock Q*
Bear	0.3	-4%	12%
Normal	0.4	14	17
Bull	0.3	26	6

What is the correlation between the two stocks?

9.5 Portfolio Standard Deviation Stock A has an expected return of 12 percent and a standard deviation of 2 percent per year. Stock B has an expected return of 15 percent and a standard deviation of 25 percent per year.

If the correlation between the returns of Stock A and Stock B is 0.4, what is the standard deviation of a portfolio comprising of A and B in equal proportions.

9.6 Beta and CAPM The risk-free rate is 8 percent and the expected return on the market portfolio is 14 percent. A portfolio that comprises the risk-free security and the market portfolio has an expected return of 11 percent and a standard deviation of 12 percent.

What is the expected return on a security that has a correlation of 0.6 with the market portfolio and a standard deviation of 25 percent?

MINICASE

You have recently graduated as a major in finance and have been hired as a financial planner by Radiant Securities, a financial services company. Your boss has assigned you the task of investing ₹ 1,000,000 for a client who has a 1-year investment horizon. You have been asked to consider only the following investment alternatives: T-bills, stock A, stock B, stock C, and market index.

The economics cell of Radiant Securities has developed the probability distribution for the state of the economy and the equity researchers of Radiant Securities have estimated the rates of return under each state of the economy. You have gathered the following information from them:

Returns on Alternative Investments

State of the Economy	*Probability*	*T-Bills*	*Stock A*	*Stock B*	*Stock C*	*Market Portfolio*
Recession	0.2	6.0%	(15.0%)	30.0%	(5.0%)	(10.0%)
Normal	0.5	6.0	20.0	5.0	15.0	16.0
Boom	0.3	6.0	40.0	(15.0)	25.0	30.0

Your client is a very curious investor who has heard a lot relating to portfolio theory and asset pricing theory. He requests you to answer the following question:

(a) What is the expected return and the standard deviation of return for stocks *A,B,C*, and the market portfolio?

(b) What is the covariance between the returns on *A* and *B*? Returns on *A* and *C*?

(c) What is the coefficient of correlation between the returns on *A* and *B*? Returns on *A* and *C*?

(d) What is the expected return and standard deviation on a portfolio in which stocks *A* and *B* are equally weighted? In which the weights assigned to stocks *A*, *B*, and *C* are 0.4, 0.4, and 0.2 respectively?

(e) The beta coefficients for the various alternatives, based on historical analysis, are as follows:

Security	*Beta*
T-bills	0.00
A	1.20
B	(0.70)
C	0.90

(i) What is the SML relationship?

(ii) What is the alpha for stocks *A*, *B*, and *C*?

(f) Suppose the following historical returns have been earned for the stock market and the stock of company *D*.

Period	*Market*	*D*
1	(5%)	(12%)
2	4	6

3	8	12
4	15	20
5	9	6

What is the beta for stock *D*? How would you interpret it?

(g) What is Capital Market Line (CML)? Security Market Line (SML)?

(h) What is the basic difference between the CAPM and the APT?

Online Resources

http://highered.mheducation.com/sites/9353166527/student_view0/chapter9/index.html

- Additional Self-Test Problems
- Additional Solved Problems
- Chapters Excel
- Excel on Solved Problems
- Answer Key

